

Question	Marking Guidance	Mark	Comments
04.1	1. Cross with homozygous recessive (fly) OR Cross with a black (fly) OR Cross with gg (fly); 2. Black offspring/fly then is heterozygous/Gg OR Black and grey offspring/fly then is Heterozygous/Gg OR No black offspring/fly then is homozygous/GG OR All grey offspring/fly then is homozygous/GG;	2 (2 x AO2)	1. Accept cross with heterozygous (fly) Alternative mark scheme, if cross not used. Mark as pairs 3 with 4, and 5 with 6. 3. <u>DNA</u> base sequencing; 4. Compare <u>base</u> sequence with known <u>alleles</u>; 5. Separate alleles using electrophoresis; 6. Use gene/DNA probes to identify alleles OR Compare position/banding with known alleles OR Homozygous forms one band, heterozygous forms two bands;

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04.2	(If sex-linked) grey/male fly would only have / pass on grey/dominant allele OR (If sex-linked) females would receive the grey/dominant allele OR (If sex-linked) grey/male fly would not have / pass on black/recessive allele OR (If sex-linked) female (offspring) would be grey OR (If sex-linked) no female (offspring) would be black	1 (1 x AO2)	Accept G for dominant allele and g for recessive allele.

	OR (If sex-linked) male (parent) could not have been heterozygous; OR (If sex-linked) only black male (parent) could produce a black bodied female;		
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04.3	1. Correct answer of 18 (%) = 2 marks;; 2. Incorrect answer but shows understanding that $2pq$ = heterozygous/carriers = 1 mark OR Incorrect answer but shows understanding that $1 - (p^2 + q^2)$ = heterozygous/carriers = 1 mark;	2 (2 x AO2)	1. Accept 0.18 for one mark. 2. Accept understanding of $2pq$ by using a calculation involving 2 x two different numbers.

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04.4	1. RrGG and Rrgg; 2. RrGg, (x2), rrGg, (and RRGg); 3. Curly(-winged), grey(-bodied) and Normal(-winged), grey(-bodied) and ratio 2 : 1;	3 (3 x AO2)	1 and 2. Accept the alleles in any order e.g. RGrG and accept if not shown on answer lines. Accept if different letters than shown are used for the alleles. 3. Accept ratios equivalent to 2 : 1. Note: If no mark awarded allow one (principle) mark when parental genotypes are incorrect but correct dihybrid genotypes shown for offspring from this cross.

